

Lecture 3. Check the quality of Linear Regression

Suppose we have built a linear regression model. Its purpose is to predict the value of y using known feature values.

For example, if we have only one feature x , the model has the form

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

If we have several features:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \cdots + \hat{\beta}_k x_{ik}$$

Here:

- i is the number of the observation;
- y_i is the actual known value for the i -th observation;
- $\hat{y}_i$ is the value predicted by the model;
- x_{ij} is the value of the j -th feature for the i -th observation;
- $\hat{\beta}_0$ is the intercept of the model;
- $\hat{\beta}_j$ is the coefficient of the j -th feature;
- k is the number of features.

The “hat” symbol $\hat{}$ means that the value has been **estimated from the available data**.

After building the model, we need to check how well it works.

Step 1. Calculate the Model Errors

First, for each observation, calculate the residual:

$$e_i = y_i - \hat{y}_i$$

Here:

- e_i is the error (residual) for the i -th observation;
- y_i is the actual value;
- $\hat{y}_i$ is the value predicted by the model.

For example, if the actual value is

$$y_i = 10,$$

and the model predicts

$$\hat{y}_i = 8,$$

then

$$e_i = 10 - 8 = 2.$$

Thus, the model made an error of 2 units.

If

$$e_i = 0,$$

then the model predicted this observation exactly.

Step 2. Calculate the Mean Squared Error (MSE)

To measure the overall error of the model, we use the mean squared error:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Here:

- MSE is the mean squared error of the model;
- n is the number of observations;
- i is the number of the observation;
- y_i is the actual value;
- $\hat{y}_i$ is the predicted value;
- $y_i - \hat{y}_i$ is the prediction error;
- $(y_i - \hat{y}_i)^2$ is the squared prediction error;
- $\sum$ means that we add the squared errors for all observations.

Thus, the procedure is very simple:

1. calculate the error for each observation;
2. square each error;
3. add all squared errors;
4. divide the result by the number of observations.

How to interpret the result:

$$\boxed{\text{MSE} = 0}$$

means that the model made no errors on any observation.

The smaller the MSE, the more accurate the model predictions are.

However, we cannot say, for example, that

$$\text{MSE} < 1$$

always means that the model is good.

MSE depends on the units and scale of y .

For example, an error of 10 dollars may be small, while an error of 10 million dollars is very large.

Therefore, MSE must always be interpreted in the context of the particular problem.

It is especially important to calculate MSE on **test data** that were not used to train the model:

$$\text{MSE}_{\text{test}} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} (y_i - \hat{y}_i)^2$$

where n_{test} is the number of observations in the test set.

Step 3. Calculate the Coefficient of Determination R^2

The next important measure is the coefficient of determination:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

First, calculate the mean value of y :

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Here:

- R^2 is the coefficient of determination;
- n is the number of observations;
- y_i is the actual value;
- $\hat{y}_i$ is the value predicted by the regression model;
- $\bar{y}$ is the mean of all y_i values;
- $\sum (y_i - \hat{y}_i)^2$ is the sum of squared model errors;
- $\sum (y_i - \bar{y})^2$ describes the total variability of y .

How to interpret R^2 :

For example,

$$R^2 = 0.9$$

means that the model explains approximately 90% of the variability in y .

If

$$R^2 \approx 1,$$

the model describes the observed values very well.

If

$$R^2 \approx 0,$$

the linear regression model provides almost no improvement over simply using the mean value $\bar{y}$.

However, it is important to remember:

high $R^2 \neq$ a guarantee of a good model.

Even a model with a high R^2 may have serious problems. Therefore, we continue the diagnostic procedure.

Step 4. Check Whether Individual Features Are Significant

For each coefficient β_j , we test the hypotheses

$$H_0 : \beta_j = 0$$

against

$$H_1 : \beta_j \neq 0.$$

Here:

- H_0 is the null hypothesis;
- H_1 is the alternative hypothesis;
- β_j is the true unknown coefficient of the j -th feature.

The hypothesis

$$H_0 : \beta_j = 0$$

means:

“There is not enough evidence to conclude that this coefficient is different from zero.”

To test this hypothesis, we use the statistic

$$Z_j = \frac{\hat{\beta}_j}{S[\hat{\beta}_j]}$$

Here:

- Z_j is the test statistic for the j -th coefficient;
- $\hat{\beta}_j$ is the estimated coefficient obtained from the sample;
- $S[\hat{\beta}_j]$ is the standard error of the coefficient estimate.

The standard error shows how uncertain the estimate $\hat{\beta}_j$ is.

Then we calculate the **p -value**.

Usually, we first choose a significance level, for example,

$$\alpha = 0.05.$$

Here α is a predefined threshold that is compared with the p -value.

If

$$p < 0.05,$$

we reject the null hypothesis:

$$\beta_j \text{ is statistically significantly different from 0.}$$

If

$$p \geq 0.05,$$

we do not have enough evidence to reject H_0 .

This **does not mean** that we have proved

$$\beta_j = 0.$$

It only means that the available data do not provide enough evidence to conclude that the coefficient is different from zero.

Step 5. Check Whether the Whole Model Is Significant

Now we test the entire regression model rather than an individual coefficient.

The null hypothesis is

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0.$$

This means that all coefficients of the features are zero.

The alternative hypothesis is

$$H_1 : \beta_1^2 + \beta_2^2 + \cdots + \beta_k^2 > 0.$$

In other words, at least one coefficient is different from zero.

We use the statistic

$$F = \frac{R^2/k}{(1 - R^2)/(n - k - 1)}.$$

Here:

- F is the test statistic used to test the significance of the whole model;
- R^2 is the coefficient of determination;
- k is the number of features;
- n is the number of observations;
- $n - k - 1$ is the number of residual degrees of freedom.

After calculating F , we determine its p -value.

If

$$p < 0.05,$$

then

$$H_0$$

is rejected.

This means that the regression model as a whole is **statistically significant**.

If

$$p \geq 0.05,$$

the available data do not provide enough evidence to confirm the statistical significance of the whole model.

Step 6. Estimate the Variability of the Errors

We estimate the error variance using

$$S_e^2 = \frac{1}{n - k - 1} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Here:

- S_e^2 is the estimate of the error variance;
- n is the number of observations;
- k is the number of features;
- $n - k - 1$ is the number of residual degrees of freedom;
- y_i is the actual value;
- $\hat{y}_i$ is the predicted value;
- $y_i - \hat{y}_i$ is the residual.

To return to the original units of y , we take the square root:

$$S_e = \sqrt{S_e^2}.$$

Here S_e is the estimated standard deviation of the errors.

The smaller S_e is, the smaller the spread of the model errors.

However, just like MSE, this value should be interpreted with respect to the scale of y .

Step 7. Check Whether the Relationship Is Really Linear

Linear regression assumes that the relationship is described by a linear function:

$$\varphi(x) = x\beta.$$

Here:

- $\varphi(x)$ is the regression function;
- x is the set of feature values;
- β is the set of regression coefficients.

The simplest way to check this assumption is to build a **residual plot**.

On one axis, we plot the predicted values

$$\hat{y}_i,$$

and on the other axis, the residuals

$$e_i = y_i - \hat{y}_i.$$

If the points are randomly scattered around the line

$$e = 0,$$

this is consistent with the assumption of linearity.

If we see a clear pattern, such as an arc or a curve, the linear model may not describe the relationship correctly.

Schematically:

random cloud of points $\Rightarrow$ a good sign

clear curved pattern $\Rightarrow$ possible non-linear relationship.

Step 8. Check Whether the Error Variance Is Constant

Ordinary linear regression assumes

$$D[\varepsilon(x)] = \sigma^2, \quad \forall x \in \mathcal{X}.$$

Here:

- $\varepsilon(x)$ is the random error of the model;
- $D[\varepsilon(x)]$ is the variance of the error;
- σ^2 is a constant value of the error variance;
- $\mathcal{X}$ is the set of possible feature values;
- $\forall x \in \mathcal{X}$ means “for every possible x ”.

In simple words:

the spread of the errors should be approximately the same for different feature values and predicted values.

This property is called **homoscedasticity**.

We can plot

$$(\hat{y}_i, e_i).$$

If the width of the cloud of residuals remains approximately constant, this is a good sign.

If the plot has a “funnel” shape:

the spread of the errors increases or decreases

then we may have **heteroscedasticity**.

For a more formal check, the lecture suggests, for example:

- White test;
- Breusch–Pagan test;
- Park test;
- Glejser test;
- Goldfeld–Quandt test.

The general idea of these tests is:

H_0 : the error variance is constant,

H_1 : the error variance is not constant.

For a significance level $\alpha = 0.05$:

$$p < 0.05 \Rightarrow \text{there is evidence of heteroscedasticity.}$$

If

$$p \geq 0.05,$$

there is not enough evidence to reject the assumption of constant error variance.

Step 9. Check the Independence of the Errors

The next assumption is that the errors of different observations should not systematically depend on each other.

For example, the error

$$e_i$$

should not automatically be large simply because the previous error

$$e_{i-1}$$

was large.

For a graphical check, we can plot the points

$$(e_{i-1}, e_i).$$

Here:

- e_i is the current residual;
- e_{i-1} is the previous residual.

This graph is called a **lag plot**.

If the points form a random cloud, this is consistent with the independence of the errors.

If there is a clear systematic pattern, the errors may be dependent.

Step 10. Check the Normality of the Errors

In classical linear regression, we assume

$$\varepsilon \sim N(0, \sigma^2).$$

Here:

- ε is the random model error;
- $\sim$ means “is distributed as”;
- N denotes the normal distribution;
- 0 is the mean of the errors;
- σ^2 is the variance of the errors.

Thus, we assume that the errors approximately follow a normal distribution centered around zero.

To check this assumption, we can use:

- a histogram of the residuals;
- a normal probability plot (Q–Q plot).

On a Q–Q plot:

points approximately follow a straight line $\Rightarrow$ the distribution is approximately normal.

If the points systematically deviate strongly from the straight line, this may indicate a violation of the normality assumption.

Step 11. Detect Outliers

An outlier is an observation that lies unusually far from the regression prediction.

First, we look at the residual

$$e_i = y_i - \hat{y}_i.$$

The larger

$$|e_i|,$$

the larger the model error for this observation.

However, ordinary residuals are not always convenient for comparing different observations.

Therefore, standardized residuals are used:

$$t_i = \frac{\hat{\varepsilon}_i}{S_e \sqrt{1 - h_i}}.$$

Here:

- t_i is the standardized residual for the i -th observation;
- $\hat{\varepsilon}_i$ is the ordinary residual;
- S_e is the estimated standard deviation of the errors;
- h_i measures how unusual the feature values of the i -th observation are;
- $\sqrt{1 - h_i}$ is a correction that takes into account the position of the observation relative to the other data.

The larger

$$|t_i|,$$

the farther the observation is from the regression line or regression surface.

The lecture also suggests the following tools for detecting problematic observations:

- box plot;
- Cook's distance;
- DFBETAS;
- DFFITS.

It is important to understand that

$$\text{outlier} \neq \text{necessarily an influential observation.}$$

An observation may lie far from the regression line but have little effect on the fitted model.

Conversely, a particular observation may strongly affect the regression coefficients.

Therefore, the influence of individual observations should be checked separately.

Step 12. Check Multicollinearity

If the model contains several features, we need to check whether some of them contain almost the same linear information.

For example, features X_1 and X_2 may be strongly linearly related.

As a first step, we can calculate their correlation:

$$r_{jl} = \text{Corr}(X_j, X_l).$$

Here:

- r_{jl} is the correlation coefficient between two features;
- X_j is the j -th feature;
- X_l is another, l -th feature;
- Corr denotes correlation.

The correlation coefficient satisfies

$$-1 \leq r_{jl} \leq 1.$$

If

$$r_{jl} \approx 1,$$

there is a strong positive linear relationship between the features.

If

$$r_{jl} \approx -1,$$

there is a strong negative linear relationship.

If

$$r_{jl} \approx 0,$$

there is no strong **linear** relationship between this pair of features.

However, it is important to remember:

$$\text{small pairwise correlations} \not\Rightarrow \text{there is definitely no multicollinearity.}$$

Multicollinearity may arise from a relationship involving several features at the same time.

Final Algorithm

After building a linear regression model, follow these steps:

1. Calculate the residuals:

$$e_i = y_i - \hat{y}_i.$$

2. Calculate the model error:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

3. Calculate the coefficient of determination:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}.$$

4. Check the statistical significance of each coefficient β_j .
5. Check the statistical significance of the whole model using the F -test.
6. Estimate the variability of the errors:

$$S_e^2 = \frac{1}{n - k - 1} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

7. Check whether the relationship can reasonably be considered linear.
8. Check whether the error variance is constant.
9. Check the independence of the errors.
10. Check the normality of the errors.
11. Detect outliers and influential observations.
12. Check the features for multicollinearity.

Very Short Summary

